Ankit Ghimire

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Research Interests

Causal Learning, Multimodal Systems, Graph Out-of-Distribution Generalization, Hyperbolic Representation Learning.

Education

University of Southern Mississippi *Aug 2023 – May 2027*

B.S. in Computer Science, Minor in Mathematics; GPA: 4.0/4.0

- Keystone Honors Scholar; merit-based full-tuition scholarship.
- **Relevant Courses:** Data Structures, Algorithms, Machine Learning in Cybersecurity, Discrete Mathematics, Linear Algebra, Probability and Statistics, Graph Theory, Real Analysis, Introduction to AI.

Peer-Reviewed Publications

A. Dahal, **A. Ghimire**, S. A. Murad, and N. Rahimi, "Redemption Score: A Multi-Modal Evaluation Framework for Image Captioning via Distributional, Perceptual, and Linguistic Signal Triangulation," *IEEE Transactions on Multimedia*, 2026. [arxiv]

A. Dahal, **A. Ghimire**, S. A. Murad, and N. Rahimi, "POVQA: Preference-Optimized Video Question Answering with Rationales for Data Efficiency," *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026. [arxiv]

Manuscripts Under Revision

A. Ghimire, S. A. Murad, and N. Rahimi, "A Depth-Aware Comparative Study of Euclidean and Hyperbolic Graph Neural Networks on Bitcoin Transaction Systems," *Under revision at IEEE Transactions on Computational Social Systems.* [arxiv]

Research Experience

Cyber Innovation Lab *Hattiesburg, MS*

Undergraduate Researcher, University of Southern Mississippi *Jan 2025 – Present*

Mentor: Prof. Nick Rahimi

Research projects:

Redemption Score: Multimodal Image Caption Evaluation

- Developed a multimodal image-caption evaluation framework combining distributional, perceptual, and linguistic signals.
- Investigated how complementary evaluation perspectives can be combined to assess generated image captions.

POVQA: Preference-Optimized Video Question Answering

- Worked on long-context multimodal video question answering with rationale supervision and preference optimization.
- Investigated rationale-guided and preference-based approaches to data-efficient video question answering.

Bitcoin Transaction Graphs: Euclidean vs. Hyperbolic GNNs

- Investigated how Euclidean and hyperbolic graph neural networks represent structure in Bitcoin transaction systems, with a focus on fraud detection and graph representation learning.
- Examined the effects of neighborhood depth, embedding geometry, and optimization on graph representation learning.

Active Intervention Selection for Causal World Models *Hattiesburg, MS*

Mentor: Prof. Bikramjit Banerjee

- Designed and implemented intervention selection algorithms for causal world models.
- Improved prediction accuracy and downstream decision-making while reducing training time compared with established baselines.

Teaching Experience

Computer Science Tutor *Hattiesburg, MS*

Computer Science Department, University of Southern Mississippi *Aug 2026 – Present*

- Tutor students in 300-level Computer Science courses, helping with course concepts and assignments.

Student Tutor *Hattiesburg, MS*

TRIO Student Support Services (TRIO-SSS) *Aug 2025 – Present*

- Provide individualized tutoring in Maths and Computer Science, adapting explanations and instructional approaches to students' learning needs.
- Contribute to academic workshops and peer mentoring initiatives that promote effective study habits and academic progress.

Honors & Awards

Second Place – Best Presentation on Computational Approaches *2026*

Undergraduate Symposium for Research and Creative Activity; Drapeau Center for Undergraduate Research ($300 award).

Keystone Honors Scholar – University of Southern Mississippi *2025 – 2027*

Selected for academic excellence and undergraduate research achievement.

Technical Skills

Programming & Tools: Python, C++, SQL, PyTorch, Pandas.

Machine Learning & Deep Learning: Logistic Regression, Random Forests, SVM, K-Means, CNNs, Transformers, GCN, GAT, GraphSAGE.